

Research Methods in Computer Science

Lecture 25: Legal, Social, Ethical and Professional Issues

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Topics

- 49 Professional Ethics
- 50 BCS Code of Conduct
- 51 ACM Code of Conduct

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Previously ...

Module aim:

To remind students of the Legal, Social, Ethical and Professional (LSEP) issues applicable to the computer industry.

47 (Social) Context

48 Ethics

- Overview
- Theories
- Relativism
- Divine
- Kantianism
- Utility
- Social contract

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Professional Ethics

- More restrictive than personal ethics because it involves the more restrictive society/culture of work and commerce
- Applies also to other restrictive social contracts such as study
- Many flavours of professional ethics exist: medical, engineering, banking, etc
- Often associated with formal structure
 - BMC (medicine), IEE (engineering), BCS (computing)
- Often formally constructed rules and codes of conduct
 - Hippocratic oath taken by doctors

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The IT Professional Context

- Legal obligations
- Social obligations
- Ownership/sharing obligations
- IP obligations
- Product development process obligations
- Product quality obligation
 - validity
 - robustness
 - simplicity
 - modifiability
 - reusability
 - compatibility
 - efficiency
 - ease of use
 - portability
 - integrity
- Product consequence obligations

The Public Interest

- Safeguarding public health and safety
- Have regards to the rights of third parties
- Understand and comply with relevant legislation
- Understand and respect issues of human rights

BCS Code of Conduct

Four component parts:

- 1 The public interest
- 2 Duty to employers and clients
- 3 Duty to the profession
- 4 Professional integrity and competence

Associated with these four components is an injunction to be aware that changes happen and change should be monitored

Duty to employers and clients

- Act professionally
- Respect the privacy and rights of employers and clients
- Be honest about products and services

Duty to profession

- Act so as to promote
 - general awareness of IT issues
 - conformance to code of conduct
- Act with integrity towards other professionals
- Do not make public statements if you are unqualified to do so

ACM Code of Conduct

- General moral imperatives
- Professional responsibilities
- Organisational leadership
- Compliance with the code

Professional competence and integrity

- Upgrade and maintain knowledge of the field
- Understand and conform to current best practices
- Avoid conflicts of interest

General moral imperatives

- Contribute to society and human well-being
- Avoid harm to others
- Be honest and trustworthy
- Be fair and take action not to discriminate
- Honor property rights including copyrights and patents
- Give proper credit for intellectual property
- Respect the privacy of others
- Honor confidentiality

Professional responsibilities

- Strive to achieve the highest quality, effectiveness and dignity in both the process and products of professional work
- Acquire and maintain professional competence
- Know and respect existing laws pertaining to professional work
- Accept and provide appropriate professional review
- Give comprehensive and thorough evaluation of computer systems and their impacts, including analysis of possible risks
- Honor contracts, agreements, and assigned responsibilities
- Improve public understanding of computing and its consequences
- Access computing and communication resources only when authorised to do so

Compliance with the code

- Uphold and promote the principles of this code
- Treat violations of this code as inconsistent with membership in the ACM

Organisational leadership

- Articulate social responsibilities of members of an organisational unit and encourage full acceptance of those responsibilities
- Manage personnel and resources to design and build information systems that enhance the quality of working life
- Acknowledge and support proper and authorised uses of an organisation's computing and communication resources
- Ensure that users and those who will be affected by a system have their needs clearly articulated during the assessment and design of requirements; later the system must be validated to meet those requirements
- Articulate and support policies that protect the dignity of users and others affected by a computing system
- Create opportunities for members of the organisation to learn the principles and limitations of computer systems